■ Live Testing Protocol: YOLO Vision System

Phase 1: Environment Replication (On Lab Machine)

Goal: Ensure the software environment matches the original setup.

• Clone/Download Repository: Download the TestYOLO_Project ZIP from GitHub and extract it.

• Setup Conda: Install Miniconda and create 'gemma_sim' environment (Python 3.10).

• Install Dependencies: pip install ultralytics opencv-python numpy

Phase 2: Software Verification (The 'Dry Run')

Goal: Confirm the AI works without hardware.

• Run Video Test: Use a downloaded video file to ensure YOLO can detect objects and the window opens correctly.

• Command: python vision_test.py --video <path_to_video>

Phase 3: Hardware Integration (The 'Real Test')

Goal: Transition from file-based simulation to live webcam.

• Connect Webcam: Plug in the USB webcam.

• Run Live Feed: Execute the script without any arguments to trigger the webcam mode.

• Command: python vision_test.py

• Observe & Validate: Check if the window opens, FPS is good, and detection works in real-time.

Summary Workflow:

Setup Machine → Verify Software (Video) → Plug Webcam → Run Live Test